Arun Vaithianathan

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Seeking a Summer 2027 internship or Spring 2027 co-op in materials R&D, process development, or scale-up for battery and energy-transition materials.

Education

Texas A&M University August 2024 – May 2028
Bachelor of Science in Chemical Engineering *College Station, TX*
- Cumulative GPA: 3.92 | Craig & Galen Brown Engineering Honors

Experience

Undergraduate Researcher, Green Lab February 2026 – Present
Artie McFerrin Department of Chemical Engineering, Texas A&M University *College Station, TX*
- Project: catalytic graphitization of petroleum coke into battery-grade anode graphite (ARPA-E, with the DOE National Energy Technology Laboratory and Oxbow Carbon). Fe_2O_3 reduces in situ to Fe as the active catalyst, with $CaCO_3$ as an additive; the target is graphite-grade crystallinity well below the 1600 °C / 24 hr industrial baseline
- Reached roughly 97% degree of graphitization at 1200 °C on a 0.5–1 hr hold across a parameter study over temperature, hold time, catalyst and additive loading, and pellet geometry; showed that $CaCO_3$ loading must track feed sulfur, the lowest collapsing crystallinity to 68%
- Execute synthesis end to end (ball milling, sieving to under 45 μm, catalyst mixing, pellet pressing, tube-furnace operation to 1600 °C, acid washing); characterized 22 samples by X-ray diffraction independently, quantifying crystallite height (L_c 419–550 Å) by (002) deconvolution
- Modeled industrial calciner residence time, sizing a 150 ft × 10 ft, 4% grade kiln at 0.59 rpm for a 2 hr target; evaluated roller compaction and briquetting as binder-free forming routes

Co-Founder January 2026 – Present
Aggie Research Finder *College Station, TX*
- Co-founded a research-discovery platform indexing 1,770+ Texas A&M faculty; grew it to 3,000+ users
- Extended it to a multi-tenant deployment across four universities with separate institutional branding, adding Rice, UT Austin, and UT Dallas

Engineering Lead, Guidance and Control Subteam May 2025 – Present
Oceanus ThinkTank, MATE ROV Competition Team, Texas A&M University *College Station, TX*
- Lead a 3-person subteam; designed the onboard architecture around 6-DOF inertial data and thruster output, validating a bench-tuned PID control loop in simulation before hardware
- Ran a single-board-computer trade study weighing cost, thermal load, and throughput against payload bandwidth requirements

Fellow and Ecosystem Builders Committee Co-Lead May 2025 – May 2026
Texas Exchange for Energy & Climate Entrepreneurship, Greentown Labs *Houston, TX*
- Assessed early-stage carbon capture, hydrogen, and low-carbon fuel technologies for process feasibility and scalability, presenting to fellows and industry practitioners
- Co-led the committee across four partner universities, organizing energy-transition events and authoring the content strategy, including live coverage from CERAWeek

Projects

XRD Degree-of-Graphitization Analyzer | *Python, SciPy, Docker*
- Built the tool computing degree of graphitization from XRD scans by Pseudo-Voigt (002) deconvolution, Bragg d-spacing, and Maire-Mering analysis, replacing the lab's manual peak-fit
- Validated to within roughly 1% of expert hand-fits; adopted by the lab as the source of its published values

Operation Halo | *Aggies Invent: ExxonMobil Engineered for Safety, 3rd Place, Team Lead* Nov 2025
- Led a team through a 48-hour design sprint on human-presence detection around heavy mobile equipment (cranes, forklifts, excavators, oilfield trucks)
- Proposed a two-stage warning and critical buffer zone from ultra-wideband hard-hat tags fused with AI cameras, tying into the equipment's existing E-stop so rental fleets need no modification
- Scoped the interlock to SIL-rated, hazardous-location service, with separate deployment cases for upstream and downstream sites

Honors & Skills

Honors: Omega Chi Epsilon (ChemE Honor Society); Gathright Dean's Excellence Award Finalist; Lechner Scholar; Dean's Honor Roll; $20K+ in scholarships; SEC Ignite Design, 2nd Place
Certifications: RPD-Research radiation-producing devices (TAMU EHS, 2026); XRD facility training (2026)
Materials: X-ray diffraction, degree-of-graphitization and crystallite-size analysis, BET, particle size distribution, tube furnace, ball milling
Process & Safety: Aspen HYSYS, HAZOP, LOPA, SIL assessment, residence-time and kiln sizing, mass and energy balances
Computational & Tools: Python, NumPy, SciPy, Pandas; Linux, Git, Docker, OriginLab, Fusion 360